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DETAIL SPECIFICATION

ROPE, FIBROUS, DOUBLE-BRAIDED, CONTINUOUS POLYESTER FILAMENT WITH A STAPLE WRAP

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fibrous double-braided polyester rope, continuous polyester filament, with a staple wrap cover and filament polyester core. The rope is furnished in one type and in the circumference specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4020

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CORDAGE INSTITUE

CI-1500A - Test Methods for Fiber Rope Physical Properties

CI-1500B - Test Methods for Fiber Rope Performance Properties

(Copies of these documents are available online at www.ropecord.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.

3.2 Materials. The inner braid shall be fabricated from bright white virgin, continuous filament, heat- and light-resistant polyester fiber. Material requirements shall be determined in accordance with 4.2.1. The outer braid shall be fabricated from white 100 percent polyester multiplex yarn.

3.3 Construction. The rope shall conform to the requirements specified herein. It shall be of double-braided construction, wherein a braid of hollow structure manufactured in a separate operation shall serve as the core, while a cover is braided over it in a second operation. In the manufacture of each braid, one-half of the yarns shall have an "S" twist, while "Z" twist shall be employed in the remaining yarns. The texture of the ultimate rope shall be firm and shall exhibit no undue sleaziness.

3.3.1 Strand configuration. All core yarns shall be one size and all cover yarns shall be one size. The number of carriers and the ends per carrier for the respective inner braid and the outer braid shall be in accordance with tables I and II. The weight of either constituent (the core or the outer braid) shall not exceed 55 percent of the total weight of the rope for any designated size.

TABLE I. Structural requirements for inner braid (core).

Cores for use in ropes of indicated circumference sizes [1/] (inches)	**Type of braid**	**Number of carriers**	**Ends per carrier**	
			Min	**Max**
¾ to 2¼ incl	Plain	8	2	6
or				
¾ to 2¼ incl	Twill	12	1	4
2½ to 5 incl	Plain	8	3	6
or				
2½ to 5 incl	Twill	12	2	4

NOTE:

[1/] These measurements shall be taken at the appropriate load "P" for the desired ultimate rope size.

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TABLE II. Structural requirements for outer braid (cover).

Rope size circumference [1/] (inches)	Type of braid	Number of carriers	Ends per carrier	
			Min	Max
¾ to 1½ incl	Twill	16 or 20	2	3
1¾ to 2½ incl	Twill	20 or 24	2	2
2¾ to 5 incl	Twill	24	2	2

NOTE:
[1/] These measurements shall be taken at the appropriate load "P" for the desired ultimate rope size.

3.4 Physical requirements. The finished rope shall conform to the physical properties specified in table III, when tested as specified in 4.2.3.

TABLE III. Physical properties.

Circumference at load "P" nominal (inches)	Diameter nominal (inches)	Load "P" (pounds)	Feet per pound at load "P"	Minimum breaking strength (pounds)
¾	¼	12	60	1,700
1	5/16	20	38	2,600
1⅛	⅜	30	26	3,600
1¼	7/16	40	19.0	4,700
1½	½	50	15	6,000
1¾	9/16	70	11	7,900
2	⅝	90	8.5	10,000
2¼	¾	110	6.7	12,200
2½	13/16	140	5.4	14,700
2¾	⅞	170	4.5	17,400
3	1	200	3.75	20,000
3¼	1 1/16	240	3.2	23,400
3½	1⅛	270	2.75	26,700
3¾	1¼	310	2.4	30,000
4	1 5/16	350	2.1	33,700
4½	1½	450	1.67	45,000
5	1⅝	550	1.35	50,000

3.4.1 Elongation. The elongation of the ropes shall not exceed 25 percent at the breaking point, when determined as specified in 4.2.3. The load elongation curve, drawn autographically, shall not exhibit evidence of changes in load applications greater than 5 percent of the load weighed at the instant of change. Changes due to splice slippage shall not be considered in this determination.

3.5 Finish. No extraneous material shall be added for the purpose of weighting the rope.

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3.6 Identification marker. The manufacturer shall identify his product by inserting an identification marker between the braids in all ropes larger than 1⅛ inches circumference. Unless otherwise specified (see 6.2), the manufacturer's name, the year of manufacture, and type of fiber (filament polyester with a staple wrap) shall be clearly printed on the marker in bold, easily-read type. Italic or script type shall not be used. The printing shall not be affected by exposure to salt water or mineral oil. This marker shall extend the entire length of the rope.

3.7 Identification tag. In addition to marker requirements specified (see 3.6), each package unit shall have a tag attached to it for identification purposes.

3.8 Put-up. The standard put-up shall be 600 foot lengths. Unless otherwise specified (see 6.2), the rope shall be furnished without knots or splices or non-returnable reels not larger than 6 feet in diameter.

3.8.1 Braider splices. Although it is desirable that no braider splices be present in the core or the cover of any size and length of rope, some methods of manufacture impose limitations. To compensate for these limitations, the following shall be considered acceptable:

3.8.1.1 Rope sizes for which braider splices shall not be permitted. For rope sizes up to and including 2¼ inches in circumference, no braider splices shall be permitted in either the core or the cover when supplied in standard put-ups (600 feet).

3.8.1.2 Rope sizes for which braider splices shall be permitted. One braider splice shall be permitted in the core and the cover for ropes whose sizes range from 2½ inches through 5 inches in circumference when supplied in standard put-ups (600 feet). In producing the splices, the distance of the overlapping shall be equivalent to eight times the circumference in inches but not less than 24 inches for ropes whose sizes are 3 inches in circumference and less.

3.8.1.3 Rope furnished with a braider splice. For any rope furnished with a braider splice, one break test in accordance with 4.2.3.4 shall be conducted with a complete braider splice in the undistributed section of the sample.

3.8.2 Workmanship. The ends of all rope shall be cut off squarely and securely whipped, taped, or heat sealed. The reels shall be wound so that each turn and layer is free from entanglement. A plus 10 percent tolerance in length will be permitted.

3.9 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 First article inspection. When specified (see 6.2), the first article sample submitted in accordance with 3.1 shall be visually inspected in accordance with table IV. The sample shall then be tested for physical and chemical requirements in accordance with 4.2.3.

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TABLE IV. Visual examination of defects.

Examine	Defect	Classification	
		Major	Minor
Appearance and workmanship	Undue sleaziness in rope texture.	X	
	Cut, any.	X	
	Chafed or damaged.	X	
	Kinks or unevenly braided, resulting in open places, breaks in continuity of braid or soft spots.	X	
	Broken, loose, or projecting ends in the core or cover.	X	
	Excessive loose fiber ends on surface.		X
	Ends not cut off squarely.		X
	Ends not securely whipped, taped or heat sealed.		X
Identification marker (Ropes 1⅛ circumference or larger)	Omitted, incorrect, illegible.	X	
	Italic or script type used.	X	
	Not completely covered by cover yarns.	X	
	Not as specified.	X	
Identification tag	Omitted, incorrect, illegible, insecurely attached.	X	
	Handwritten entries.		X
Cleanliness	Spot or stain, clearly visible. 1/		X

NOTE:
1/ At normal inspection distance (approximately 3 feet).

4.2 Inspection. Sampling and inspection shall be performed in accordance with MIL-STD-1916, except where otherwise indicated.

4.2.1 Component inspection. Determination shall be made for conformance with all characteristics of table V.

TABLE V. Component test.

Characteristic	Requirement	Test method
Yarn sizes a. Single size for core. b. Single size for cover.	3.3.1	Visual
Braider splices in core	3.8.1	Visual
Number of carriers – inner braid Number of ends/carrier – inner braid	Table I	Visual
Number of ends/carrier – outer braid Number of carriers – outer braid	Table II	Visual

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4.2.2 Examination of the end item for visual defects. The defects listed in table IV shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The unit of product for this examination shall be one reel. Ten percent of the gross length contained on each sample unit, but not less than 100 feet, shall be subjected to the visual examination. Examination of the end item for visual defects shall be in accordance with MIL-STD-1916.

4.2.2.1 Defects with regard to length. Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

a. Length of unit less than or more than length specified (including permitted tolerances).

b. Length of unit less than marked on tag.

c. Reels not in continuous length.

4.2.2.2 Defects with regard to winding. Defects with regard to winding shall be considered to exist if the following are determined during inspection:

a. Improperly or not firmly wound resulting in slippage during unwinding or otherwise affecting free unhampered unwinding of rope.

b. Knot, splice, or otherwise joining of ends to make a continuous length.

4.2.3 Test methods. The methods of testing shall be as specified in table VI.

TABLE VI. Test methods.

Characteristic	Requirement	Test method	Number of determinations per individual sample unit 1/	Results reported as
No. of carriers (cover and core)	Table I and II	Visual	1	Pass or fail
No. of ends per carrier (cover and core)	Table I and II	Visual	1	Pass or fail
Circumference	3.4	4.2.3.2	3	Pass or fail
Linear density	3.4	4.2.3.3	1	Pass or fail
Breaking strength	3.4	4.2.3.4	3	Pass or fail
Elongation	3.4.1	4.2.3.5	1	Pass or fail

NOTE:

1/ A random sample of reels shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and acceptance quality limit (AQL), as specified (see 6.2).

4.2.3.1 Specimen preparation. The designated length and number of test specimens shall be removed from selected test reels.

4.2.3.1.1 Length and weight. The length measurement shall be made in accordance with CI-1500A.

4.2.3.1.2 Buried eye splices. The specimens shall then be prepared with buried eye splices.

4.2.3.2 Circumference and picks per inch.

4.2.3.2.1 Circumference. The circumference measurement shall be made in accordance with CI-1500A.

4.2.3.2.2 Picks per inch. Picks per inch shall be determined in accordance with CI-1500A.

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4.2.3.3 Linear density. Linear density (pounds per 100 feet) will be determined in accordance with CI-1500A.

4.2.3.4 Breaking strength. Breaking strength shall be determined in accordance with CI-1500B.

4.2.3.4.1 Rope furnished with a braider splice. If any rope is furnished with a braider splice, one break test shall be conducted with a complete braider splice in the undistributed section of the same. Failure of the sample to meet the minimum breaking strength of table III is cause for rejection of the lot.

4.2.3.4.2 Braider splice appearance. Any braider splice appearing too unsightly in the opinion of the Government representative may be removed from the rope and tested in accordance with 4.2.3.4.1.

4.2.3.5 Elongation. Elongation shall be determined in accordance with CI-1500B.

4.2.3.5.1 Autographic recording. The autographic recording obtained for each specimen while conducting the breaking strength test may be employed as an alternate method in determining the breaking elongation when industrial safety regulations will not permit use of the method described above.

4.2.3.6 Manufacturer's certificate. Because splices within the core are difficult to detect subsequent to over application of the cover, a manufacturer's certificate attesting to the number of such splices shall be considered acceptable for this situation. However, a record of verifiable information shall be available to the inspector.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The rope covered by this specification is intended for use in high line personnel transfer between naval ships and other application that require maximum gripping surface and minimum elongation, while maintaining maximum breaking strength.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Circumference (size) required (see 1.1 and table III).

c. When first article sample is required (see 3.1 and 4.1).

d. When information on identification marker is other than specified (see 3.6).

e. Put-up and length when other than specified (see 3.8).

f. AQL of 1.0 (see table VI).

g. Selection of applicable levels of packaging and packing (see 5.1).

h. That purchaser will accept at original weight, any unit which has been shortened or cut for test specimens, if in complete compliance with this specification.

i. That this rope will be purchased on a price-per-pound basis (net weight).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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6.4 Use of existing stocks. Current stocks of materials may be used until depleted; future requisitions for material should refer to the current version of this specification.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.6 Subject term (key word) listing.

Breaking strength

Elongation

Splice

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CQ

Preparing activity:
Navy – SH
(Project 4020-2017-004)

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